



KZY-EDM

Electroconductive Zirconia Composite

KEY PROPERTIES

- Electrically conductive ceramic
- Suitable for EDM machining
- High strength
- Excellent surface finishing
- Low friction against metals
- Excellent chemical resistance

APPLICATIONS

- Cold forming tools
- Tooling components
- Precision wear parts

MAIN PROPERTIES

Property	Value	Unit
Density	5.60	g/cm ³
Flexural strength	1000	MPa
Fracture Toughness	6.0	MPa·√m
Hardness	14	GPa
Young Modulus	220	GPa
Electrical Resistivity	3×10 ⁻⁵	Ω·cm

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)	900	°C
Recommended temp	<850	°C
Continuous temp	800	°C

ELECTRICAL BEHAVIOR

Conductive	
EDM Capability	Yes

MATERIAL INFORMATION

Material type	Conductive zirconia composite
Structure	SPS sintered ceramic
Key feature	EDM machinability
Machinability	EDM
Color	Dark grey / Black

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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